

🌸 Understanding Arrays & Lists 🌸



What is an Array?

An **array** is like a container that holds multiple values in a specific order. Think of it as a list!

Why use arrays?

- Store multiple related items together
- Keep data organized
- Easy to loop through items
- Access items by their position (index)

Real-life example:

Imagine a student roster:

```
Position 0: Ana  
Position 1: Maria  
Position 2: Sofia  
Position 3: Emma
```

Important: Arrays start counting at **0**, not 1!

✨ Creating Arrays

Method 1: Using brackets []

```
let alunas = ["Ana", "Maria", "Sofia"];  
let numbers = [1, 2, 3, 4, 5];  
let mixed = ["Emma", 18, true];
```

Method 2: Array of Objects

You can create arrays that hold objects with multiple properties:

```
let alunas = [  
  {  
    nome: "Ana",  
    saborSorvete: "Chocolate"  
  },  
  {  
    nome: "Maria",  
    saborSorvete: "Strawberry"  
  },  
  {  
    nome: "Sofia",  
    saborSorvete: "Vanilla"  
  }  
];
```

💡 **Tip:** Each object in the array can have multiple properties (nome, idade, favorite things, etc.)

Accessing Array Items

Using Index Numbers

```
let alunas = ["Ana", "Maria", "Sofia"];

console.log(alunas[0]); // Ana
console.log(alunas[1]); // Maria
console.log(alunas[2]); // Sofia
```

Accessing Object Properties

```
let alunas = [
  { nome: "Ana", saborSorvete: "Chocolate" },
  { nome: "Maria", saborSorvete: "Strawberry" }
];

console.log(alunas[0].nome);
// Output: Ana

console.log(alunas[0].saborSorvete);
// Output: Chocolate
```

Common Mistakes ❌

- Forgetting arrays start at 0
- Trying to access index that doesn't exist
- Mixing up dot notation (.) for objects



Looping Through Arrays

For Loop - Classic Method

```
let alunas = ["Ana", "Maria", "Sofia"];

for (let i = 0; i < alunas.length; i++) {
  console.log(alunas[i]);
}
```

```
Ana Maria Sofia
```

For...of Loop - Easier Way!

```
let alunas = ["Ana", "Maria", "Sofia"];

for (let aluna of alunas) {
  console.log(aluna);
}
```

```
Ana Maria Sofia
```



Exercise Solution

Task: You have a list of students (alunas), and each one has a name (nome) and favorite ice cream flavor (saborSorvete). Write a program that loops through this list and prints the name and corresponding favorite ice cream flavor.

Complete Solution with Template Literals:

```
let alunas = [
  { nome: "Ana", saborSorvete: "Chocolate" },
  { nome: "Maria", saborSorvete: "Strawberry" },
  { nome: "Sofia", saborSorvete: "Vanilla" }
];

for (let i = 0; i < alunas.length; i++) {
  console.log(`Nome: ${alunas[i].nome} | Sabor favorito: ${alunas[i].saborSorvete}`);
}
```

```
Nome: Ana | Sabor favorito: Chocolate Nome: Maria |
Sabor favorito: Strawberry Nome: Sofia | Sabor favorito:
Vanilla
```

Alternative: Using for...of

```
for (let aluna of alunas) {
  console.log(`Nome: ${aluna.nome} | Sabor favorito: ${aluna.saborSorvete}`);
}
```

Understanding the Syntax:

- `alunas[i]` - accesses the object at position `i`
- `alunas[i].nome` - gets the "nome" property from that object
- `${alunas[i].nome}` - template literal syntax to insert variable value
- ``backticks`` - used for template literals (not regular quotes)



Console.log Tips

What is console.log?

It's a way to print/display information in the browser's console (developer tools).

Different Ways to Print:

Concatenation with +

```
let nome = "Ana";  
console.log("Nome: " + nome);
```

```
Nome: Ana
```

Template Literals with \${ }

```
let nome = "Ana";  
console.log(`Nome: ${nome}`);
```

```
Nome: Ana
```



How to see console.log:

1. Open your browser
2. Press F12 (or right-click → Inspect)
3. Click on "Console" tab
4. Run your code and see the output!